

Elements of Position

III. Without a Landing

The residual the operators leave, and what the pair still forces

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A note. Part II named an entrance residual and then, in places, treated it as dirt on an idealization. This part writes the residual as an object. A heading identity is checked against u_m . A pair identity is checked against m , by Part I. Plates L1 and L2 read the recorded tables. They are not a second walk.

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The leftover of the landing is forced

Hit m occurs at step $j_m = \lceil m\lambda \rceil$ whenever $m\lambda$ is not an integer, $\lambda = \pi / \arctan h$. The heading has already passed $m\pi$ by

$$\varepsilon_m = \arctan(h) (1 - \{m\lambda\}).$$

Proposition 1. $u_m = ((-1)^m \cos \varepsilon_m, (-1)^m \sin \varepsilon_m)$.

On the recorded kick the first residual is the vertical of the first hit: $h_y = -0.00040316$ at $m = 1$. The continuous-clock detector that waits for $y > 0.05$ and then $y \leq 0$ arrives at the same heading. The operators do not contain the wall. A hit is the first sample after the crossing.

Proposition 2. $\varepsilon_m = 0$ if and only if $m\lambda \in \mathbb{Z}$.

That is the half-turn mix of Part II, and only that mix.

What does not need a landing

These are identities of the pair and of inversion. They use u_m only as a ray.

$$|OP_m| |OQ_m| = 1, \quad Q_m = P_m / |P_m|^2, \quad \left(\frac{1}{m}, m; 1, -1\right) = -1, \quad \tau(m) = \frac{(m-1)^2}{2m}.$$

The circle on diameter OP_m meets the unit circle at points whose foot on the ray is exactly Q_m . Foot error on the recorded headings: 2×10^{-16} . The mouth of that kite is $2\sqrt{1-1/m^2}$. At the lock the mouth is 0. At the first split it is $\sqrt{3}$. It walks toward 2 and does not reach 2 at any finite hit.

Opposite-wall chords $|P_m - P_{m-1}| = 2m - 1$ assume $u_m = -u_{m-1}$. The operators deliver $u_m \cdot u_{m-1} = -\cos(\varepsilon_m - \varepsilon_{m-1})$. The missing length is of order ε^2 .

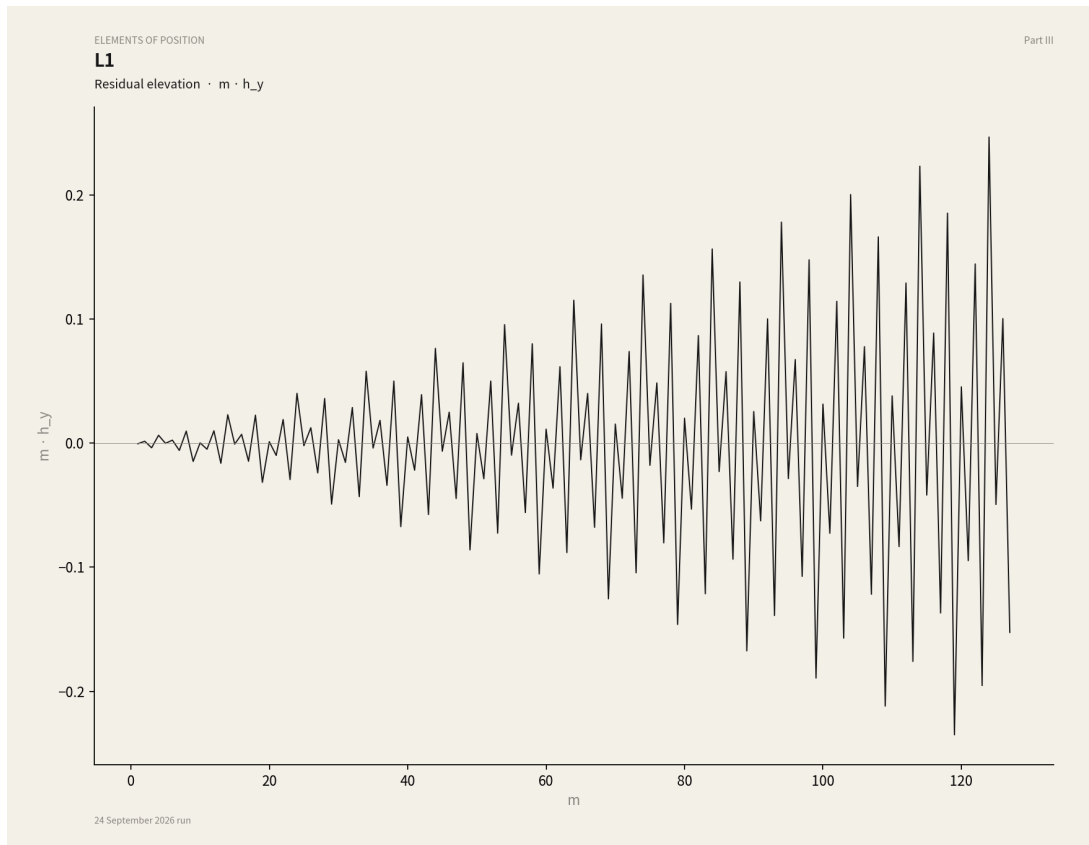


Figure 1: Sheet L1. Elevation of $m \cdot h_y$ on the recorded hits. The residual is the object.

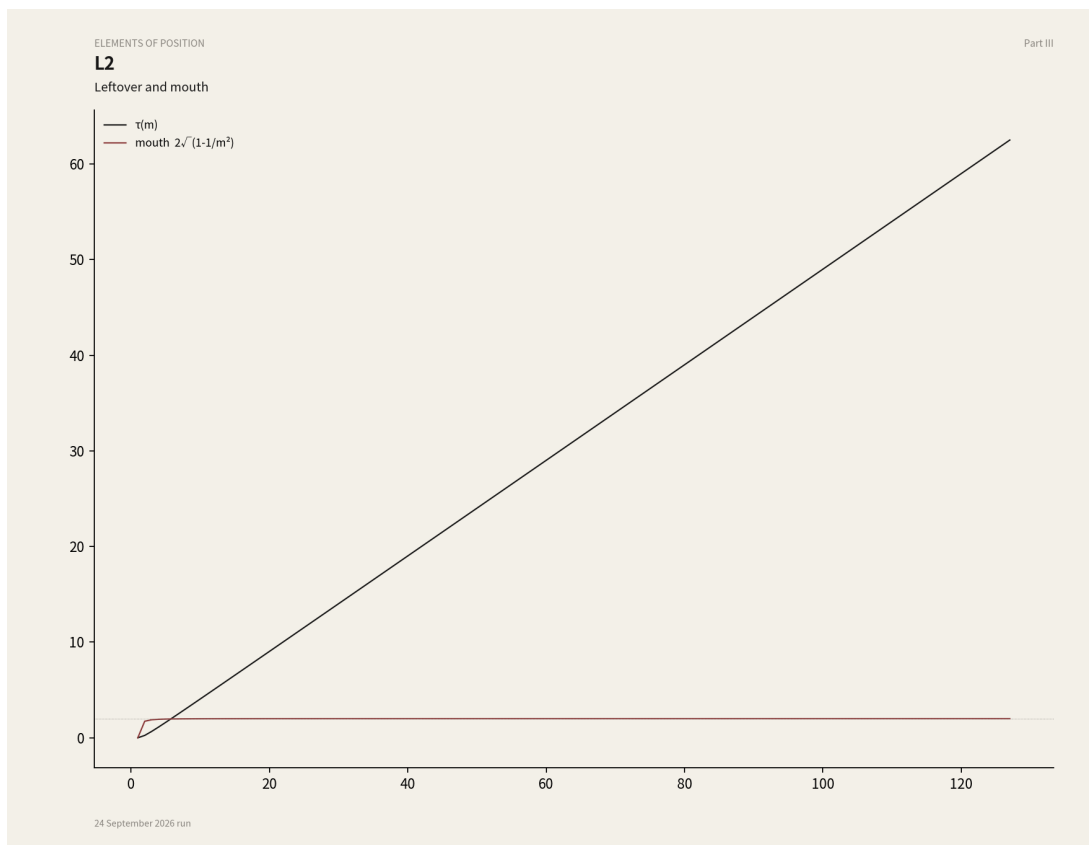


Figure 2: Sheet L2. Leftover climbs. Mouth jumps at the first split and then sits toward 2.

The kite factor

At every hit the pair writes $1 - 1/m^2$. That is the square of half the mouth, and also $1 - |Q_m|^2$.

$$\prod_{m=2}^N \left(1 - \frac{1}{m^2}\right) = \frac{N+1}{2N}.$$

It walks toward $1/2$. Integers only.

Kept only at new inners — hits whose inner $1/m$ is not a product of earlier inners — the same product walks toward a named comparison $6/\pi^2$. The name is a sheet reading. The object is the product of kite factors at those hits. New inner is primality of the hit index, named as that, not as a narrower “full web.”

The mix family

$T_{a,b}(x, y) = (x - ay, y + bx)$, then bind. Opposite signs return. Equal $a = b$ is balance, not a new kind. Same-sign Euclidean bind does not return. The minus is the kind. The residual law rides on whatever kick still returns.

What stands

Forced at every residual: the heading formula; the pair, the inversion, the kite-foot, the right angle at X ; exact closure exactly when $m\lambda$ is an integer, and only on the half-turn mix.

Forced only as well as two headings agree: opposite-wall chords, their product, the swap of collinear diameters.

Not granted: a landing on the cut; α of a first block treated as a constant of the operators; a value of 4, or of $1/2$, or of 2, treated as reached at a finite hit.

Where is 1? It is the lock. Even there the operators did not land. They only met themselves.